

# Beyond PII Removal: Risk-Budgeted Generalization for LLM-Ready Sensitive Text

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## Abstract

1 Sensitive text corpora are valuable for retrieval, evaluation, and post-  
 2 training adaptation of foundation-model systems, but common de-  
 3 identification pipelines still center on direct personally identifying infor-  
 4 mation (PII): names, emails, phone numbers, addresses, and account num-  
 5 bers. This is necessary but incomplete. A record with no explicit name  
 6 can still reveal sensitive attributes through quasi-identifiers such as exact  
 7 dates, fine-grained locations, rare occupations, institutions, family relations,  
 8 clinical or legal events, citizenship cues, and combinations of ordinary  
 9 facts. We introduce Risk-Budgeted Quasi-Identifier Generalization (RB-  
 10 QIG), a lightweight transformation policy that treats de-identification as a  
 11 privacy–utility allocation problem: given candidate quasi-identifiers, which  
 12 facts should be kept, generalized, or redacted under a document-level risk  
 13 budget? RB-QIG uses transparent risk and utility scores, span-level gen-  
 14 eralization ladders, and a greedy audit log rather than model training or  
 15 unconstrained rewriting. Across synthetic, RAT-Bench, and TAB pilots, the  
 16 central pattern is consistent: direct PII redaction and naive LLM sanitiza-  
 17 tion leave substantial residual inference risk, while explicit quasi-identifier  
 18 transformation sharply reduces measured leakage. The strongest utility  
 19 result is controlled: on synthetic records, RB-QIG preserves far more utility  
 20 facts than blanket quasi-identifier redaction at a nonzero but much lower  
 21 leakage level. Public utility is more mixed, and stronger GPT-5.5 attack-  
 22 ers recover more from all methods, so RB-QIG should not be read as an  
 23 anonymization guarantee. Instead, the result is a practical argument for  
 24 evaluating what capable models can still infer, not only which explicit PII  
 25 spans were removed.

## 1 Introduction

27 Sensitive text is one of the most tempting and most difficult data sources for foundation-  
 28 model systems. Clinical notes, legal opinions, financial communications, student-support  
 29 records, workplace logs, and user-assistant conversations contain the domain knowledge  
 30 that makes retrieval and adaptation useful. They also contain facts that can identify, profile,  
 31 or expose people. Responsible data enablement therefore often starts with a familiar  
 32 transformation step: remove names, emails, phone numbers, addresses, and other direct  
 33 identifiers.

34 This paper asks what remains after that step. A sentence such as “a 34-year-old oncology  
 35 infusion pharmacist at a small startup in Asheville called about a brother’s rare condition  
 36 after an April 2025 surgery” can be name-free and still highly revealing. The risk is not a  
 37 single missed token; it is the combination of age, job, location, relationship, date, and rare  
 38 event. Classical privacy work formalized this linkage problem through quasi-identifiers and  
 39 group-based anonymization [Sweeney \(2002\)](#); [Machanavajjhala et al. \(2007\)](#); [Li et al. \(2007\)](#);  
 40 [Fung et al. \(2010\)](#); [Narayanan & Shmatikov \(2008\)](#). Regulatory and policy guidance also  
 41 recognizes that identifiability can arise directly or indirectly, and that residual information  
 42 may be identifying in combination with other available information [European Parliament](#)  
 43 [and Council of the European Union \(2016\)](#); [U.S. Department of Health and Human Services](#)  
 44 [\(2026\)](#); [McCallister et al. \(2010\)](#).

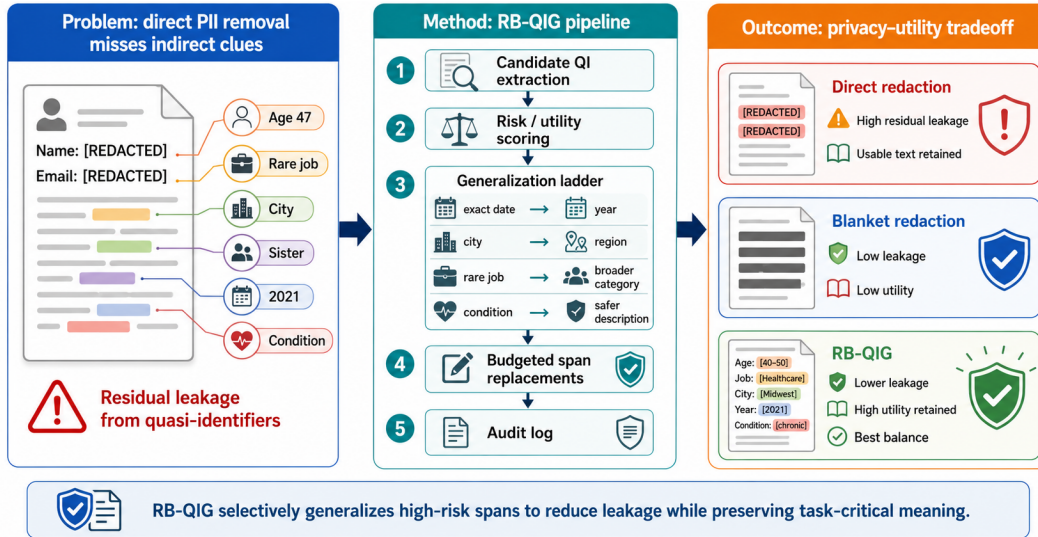


Figure 1: Conceptual overview of RB-QIG. Direct PII removal can leave indirect quasi-identifier clues; RB-QIG extracts candidate QIs, scores privacy risk and utility, and applies budgeted generalization so high-risk spans are coarsened rather than always left intact or blanket-redacted.

Large language models make this residual-risk problem more visible. They can combine paraphrases, world knowledge, and conversational cues to infer private attributes that are not explicitly stated [Staab et al. \(2024a\)](#). Recent work on semantic privacy and implicit inference argues that users and systems often underestimate what can be inferred from ordinary text [Ma et al. \(2025\)](#); [Wang et al. \(2025\)](#). Text anonymization benchmarks such as TAB and RAT-Bench likewise move beyond direct PII spans toward residual re-identification risk and utility preservation [Pilán et al. \(2022\)](#); [Krčo et al. \(2026\)](#).

The practical design question is therefore not “did a recognizer remove every PII span?” but “what can an attacker still infer, and how much useful meaning survives the transformation?” Direct PII removal is utility-preserving but leaky. Blanket quasi-identifier redaction is safer but often destroys the semantics that make the data useful. We study the middle ground: targeted generalization. For example, “oncology infusion pharmacist in Asheville” can become “clinical pharmacy professional in a city in the southeastern U.S.” rather than either being left intact or replaced with opaque placeholders.

We introduce Risk-Budgeted Quasi-Identifier Generalization (RB-QIG), a small auditable policy layer for sensitive text. RB-QIG does not train a model and does not rewrite whole documents. It assumes candidate quasi-identifier spans from an oracle, benchmark-aware extractor, or blind extractor; assigns each candidate a privacy-risk score and a utility-importance score; and applies the least destructive replacement needed to meet a document-level risk budget. Each edit is logged, making the policy inspectable by data owners.

65 Our contributions are:

- a taxonomy and transformation view of quasi-identifiers for LLM-ready sensitive text;
- a risk-budgeted span-generalization algorithm that exposes a tunable privacy-utility frontier without fine-tuning;
- an evaluation protocol combining deterministic leakage metrics, LLM attackers, stronger-attacker checks, and utility diagnostics; and
- an empirical analysis showing that direct redaction fails because sensitive attributes remain inferable from relational, lexical, and contextual cues.

## 2 Background

**Direct identifiers, quasi-identifiers, and sensitive attributes.** We use *direct identifier* for a span that directly identifies a person or account: a name, email, phone number, address, account ID, or equivalent. We use *quasi-identifier* (QI) for a fact that may not identify a person alone but becomes identifying or sensitive in combination with other facts. Examples include ages, dates, locations, rare occupations, schools, employers, family relations, medical events, legal events, citizenship status, and cultural cues. We use *sensitive attribute* for the attribute an attacker tries to recover, such as sex, race/ethnicity, citizenship, marital status, education, employment, or a domain-specific clinical/legal fact.

**Why text is harder than tables.** Classical anonymization methods were developed largely for structured records, where each row has known columns and generalization can be specified over attribute hierarchies Sweeney (2002); Machanavajjhala et al. (2007); Li et al. (2007). Text does not offer that clean schema. The same cue may appear as a named entity, a narrative phrase, a family relation, a quoted statement, or a placeholder created by a previous sanitizer. Recent medical and general text work has therefore emphasized indirect identifiers and LLM-based inference risks Baroud et al. (2025); Jiang et al. (2026); Staab et al. (2024b); Yang et al. (2024).

**Threat model.** The attacker receives transformed text. In our LLM-attacker evaluations, the attacker is asked to infer benchmark attributes and return structured guesses with short evidence. We score exact recovery, coarse recovery, record compromise, and risk-weighted leakage. The attacker is not asked to identify real people, and all experiments use synthetic or public benchmark data. This is not a complete model of arbitrary external databases or determined human investigators. It is a reproducible stress test for whether a transformation leaves sensitive attributes inferable from the transformed text itself.

**Three evaluation settings.** We separate transformation quality from extraction realism. In the *controlled* setting, oracle quasi-identifier spans are known from synthetic profiles. In the *target-aware public* setting, benchmark target attributes are used to find value-revealing spans, isolating how well transformations work when the sensitive attributes of interest are known. In the *blind public* setting, the extractor does not see target attributes at extraction time and is scored afterward. The blind setting is closest to deployment and is the main source of our limitations.

## 3 Risk-Budgeted Quasi-Identifier Generalization

### 3.1 Conceptual Design

RB-QIG treats de-identification as a budgeted editing problem. For each candidate quasi-identifier  $q_i$ , the system stores: (1) its text span; (2) a type such as AGE, LOCATION, OCCUPATION, or FAMILY; (3) a privacy-risk score; (4) a utility-importance score; and (5) a short ladder of possible replacements. Table 1 shows the main QI types.

The important distinction is between *redaction* and *generalization*. Redaction replaces a value with a placeholder such as [LOCATION]. Generalization keeps a coarser version of the useful semantics, such as “city in the southeastern U.S.”. RB-QIG’s hypothesis is that many sensitive-data uses do not require exact quasi-identifiers; they require enough contextual information for the downstream task.

### 3.2 Scoring and Budgeted Editing

Each candidate  $q_i$  receives a risk score  $r_i \in [0, 5]$  and utility score  $u_i \in [0, 5]$ . Risk reflects specificity, rarity, sensitivity, and linkability. Utility reflects whether the fact is useful for the benign task. For instance, a diagnosis family may be useful in a medical-support note, whereas the exact surgery date may not be.

Table 1: Examples of quasi-identifier categories and generalization ladders. The method uses span replacements rather than unconstrained rewriting, so every edit can be audited.

| Category            | Precise cue                        | Generalized cue                    |
|---------------------|------------------------------------|------------------------------------|
| Date/age            | April 17, 2025; 34                 | month/year; 30s                    |
| Location            | Asheville, NC                      | city in the southeastern U.S.      |
| Occupation          | oncology infusion pharmacist       | clinical pharmacy professional     |
| Institution         | named school or employer           | university; healthcare company     |
| Medical/legal event | rare condition or case event       | broadier condition/event type      |
| Family relation     | younger brother under guardianship | dependent family member            |
| Demographic cue     | citizenship or heritage phrase     | broadier demographic phrase        |
| Narrative cue       | unusual life event                 | coarse role-preserving description |

121 The document risk score is

$$R(x) = \sum_i r_i + 0.5 \binom{|\{i : r_i \geq 3\}|}{2},$$

122 where the second term is a simple combination-risk penalty: multiple high-specificity cues  
 123 are more dangerous together than separately. Each candidate replacement has an estimated  
 124 risk drop and utility cost. RB-QIG greedily applies the next replacement with the best  
 125 risk-drop-per-utility-loss ratio until the risk budget is met or no replacement remains. The  
 126 output is the transformed text plus an edit log.

127 This scoring rule is deliberately simple. It is not a universal privacy estimate. Its role is to  
 128 make a data owner’s policy explicit: which facts are risky, which facts are useful, how much  
 129 residual risk is tolerated, and which semantic abstractions are acceptable.

### 130 3.3 Baselines

131 We compare RB-QIG against conceptual and practical baselines: *none* (unchanged text),  
 132 *direct redaction* (direct identifiers only), *naïve LLM direct* (an LLM instructed to remove  
 133 identifying information while preserving meaning), *blanket QI redaction* (all QIs become  
 134 typed placeholders), and a *Presidio-style pattern baseline*. The pattern baseline is a practical  
 135 lower-bound diagnostic, not a full production Presidio deployment.

## 136 4 Experiments

137 **Datasets.** The controlled benchmark contains 100 synthetic records across medical, legal,  
 138 financial, workplace, and education-support domains. Each record has direct identifiers,  
 139 QIs, target attributes, a task label, and utility facts. The public text benchmark is a 100-record  
 140 English difficulty-1 RAT-Bench pilot Krčo et al. (2026). We also run a 30-document TAB  
 141 ECHR legal screen Pilán et al. (2022). These public datasets are used as stress tests rather  
 142 than claims of benchmark saturation.

143 **Evaluators.** Deterministic evaluators score explicit attribute leakage, QI specificity, and  
 144 token change. LLM attackers score whether target attributes can be recovered from trans-  
 145 formed text. LLM utility judges score label preservation, fact preservation, and semantic  
 146 utility without rewarding preserved direct identifiers. Additional bounded checks use  
 147 GPT-5.5 as a stronger attacker and a mini-model agreement screen to test model dependence.  
 148 All API calls use synthetic or public benchmark records.

149 **Metrics.** We report risk-weighted leakage as the primary privacy metric. Utility is mea-  
 150 sured with utility-fact preservation in the controlled setting, semantic utility in LLM-judge  
 151 settings, and annotation-derived QI specificity as a narrow no-API proxy for whether  
 152 typed/generalized semantics survive. To keep the main paper readable, the results section  
 153 emphasizes headline patterns; the appendix contains the detailed numeric tables.

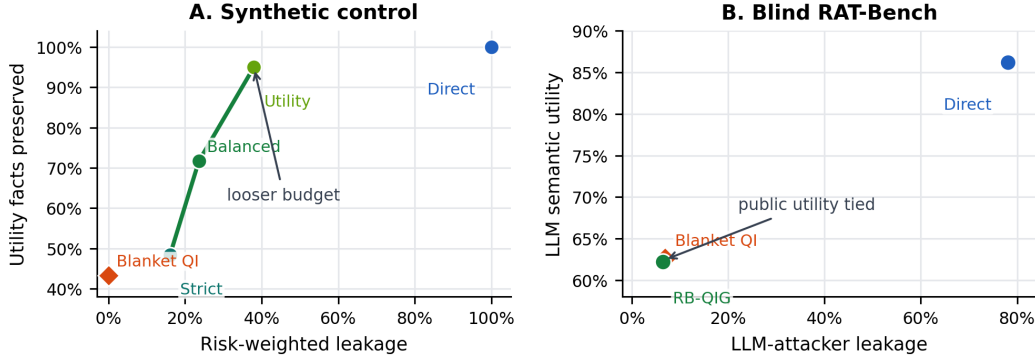


Figure 2: Privacy-utility frontier. Panel A uses deterministic synthetic metrics: direct redaction preserves utility facts but leaks QIs, blanket QI redaction removes measured leakage with lower utility, and RB-QIG budgets trace an intermediate frontier. Panel B uses blind RAT-Bench LLM attacker and utility judges; RB-QIG and blanket redaction are close on public utility.

Table 2: Main results.

| Question                                          | Main observation                                                               | Interpretation                                                              |
|---------------------------------------------------|--------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| Does direct PII redaction suffice?                | No. RAT-Bench direct redaction leaves high LLM-attacker leakage.               | Sensitive attributes remain inferable from QIs and context.                 |
| Can RB-QIG reduce residual risk?                  | Yes. RB-QIG sharply lowers leakage relative to direct and naive LLM redaction. | Explicit QI transformation is necessary for LLM-ready text.                 |
| Is RB-QIG more private than blanket QI redaction? | No clear evidence. They are usually tied under LLM attackers.                  | The value of RB-QIG is utility-aware transformation, not privacy dominance. |
| Where is the utility win?                         | Strong in synthetic utility facts; weak or tied in public utility judges.      | Utility-preserving blind rewriting remains the main open problem.           |
| Do stronger attackers change the conclusion?      | GPT-5.5 increases absolute leakage, but the method ordering remains.           | Residual risk is real; robust evaluation needs strong attackers.            |

## 5 Results

**Result 1: direct PII redaction is the wrong endpoint.** Across the RAT-Bench pilots, removing direct identifiers leaves substantial recoverable information. The target-aware LLM attacker still recovers many attributes after direct redaction, and the blind public setting shows the same qualitative failure. A naive LLM sanitizer improves over direct redaction but does not close the gap. The takeaway is simple: for LLM-ready sensitive text, PII removal is a first pass, not a complete transformation.

**Result 2: RB-QIG gives a controlled middle ground.** The cleanest privacy-utility result is the synthetic control (Figure 2). Direct redaction preserves the utility facts but leaves QI leakage. Blanket QI redaction removes measured leakage but discards much of the useful semantics. RB-QIG balanced sits between these extremes: it greatly reduces leakage while preserving substantially more utility facts than blanket redaction. This is the paper’s strongest positive utility claim, and it should stay scoped to the controlled setting.

**Result 3: public utility is the caveat, not the headline.** On blind public RAT-Bench, RB-QIG and blanket redaction are similar under the LLM utility judge. Additional placeholder,

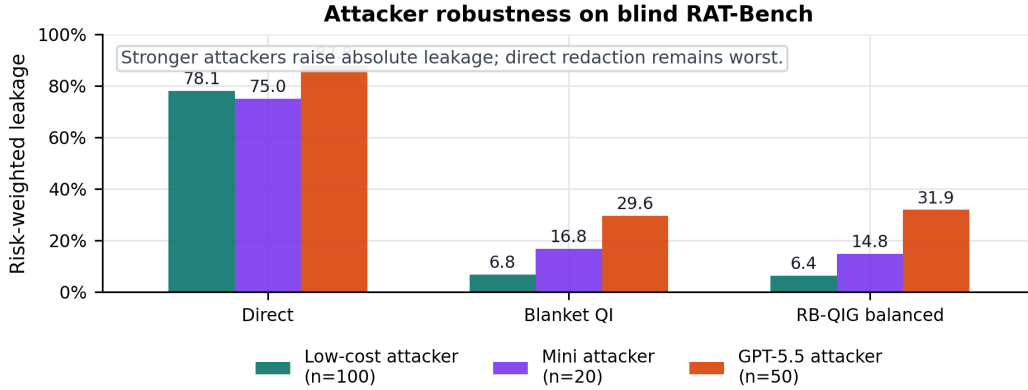


Figure 3: Attacker robustness on blind RAT-Bench. Bars are for low-cost, mini, and GPT-5.5 attackers with sample sizes shown in the legend. Stronger attackers raise absolute leakage, but direct redaction remains much leakier than RB-QIG or blanket QI redaction.

169 fluent-rewrite, and attribute-level fluent diagnostics did not convert this into a clear public  
 170 utility win. The text-quality audit suggests that surface fluency alone is not the missing  
 171 ingredient: cleaner phrase-level rewrites can still remove the facts needed for hidden-label  
 172 utility. This points toward a more difficult future direction: task-aware generalization rather  
 173 than prettier replacement phrases.

174 **Result 4: stronger attackers raise leakage but preserve the ordering.** A bounded GPT-  
 175 5.5 attacker check is harsher than the low-cost attacker: absolute leakage rises for all  
 176 privacy-preserving methods. However, direct redaction remains much leakier than RB-QIG  
 177 or blanket QI redaction, and a mini-model agreement check gives the same qualitative  
 178 ordering. This supports the use of strong attackers as robustness tests, while reinforcing  
 179 that RB-QIG is not an anonymization guarantee.

180 **Result 5: the risk budget behaves like a real control knob.** Budget-frontier diagnostics  
 181 show the expected behavior on blind RAT-Bench and TAB: stricter budgets reduce leakage  
 182 but also reduce annotation-derived specificity, while looser budgets preserve more typed or  
 183 generalized QI semantics at higher residual risk. This is important because RB-QIG is meant  
 184 to be a policy interface. Different data owners may choose different budgets depending on  
 185 the recipient, task, and review process.

186 **Cross-domain and practical-baseline diagnostics.** The TAB legal screen supports the  
 187 residual-risk story in a second domain: direct redaction leaves many legal and narrative  
 188 cues, while RB-QIG reduces deterministic leakage. However, the bounded legal-task utility  
 189 screen is tied or slightly worse than blanket redaction, so TAB should be presented as  
 190 cross-domain privacy evidence rather than a utility win. The Presidio-style pattern baseline  
 191 provides a practical lower-bound diagnostic: pattern-only redaction misses many domain-  
 192 specific identifiers, especially legal names and case identifiers.

193 **Failure modes.** Residual leaks usually come from relational or lexical cues rather than  
 194 missed obvious spans: bereavement language can imply marital status, gendered family  
 195 roles can imply sex, cultural references can imply race or ethnicity, citizenship phrasing  
 196 can reveal immigration status, and institutions can reveal education or employment. These  
 197 failures explain why span recall alone is an incomplete privacy metric.

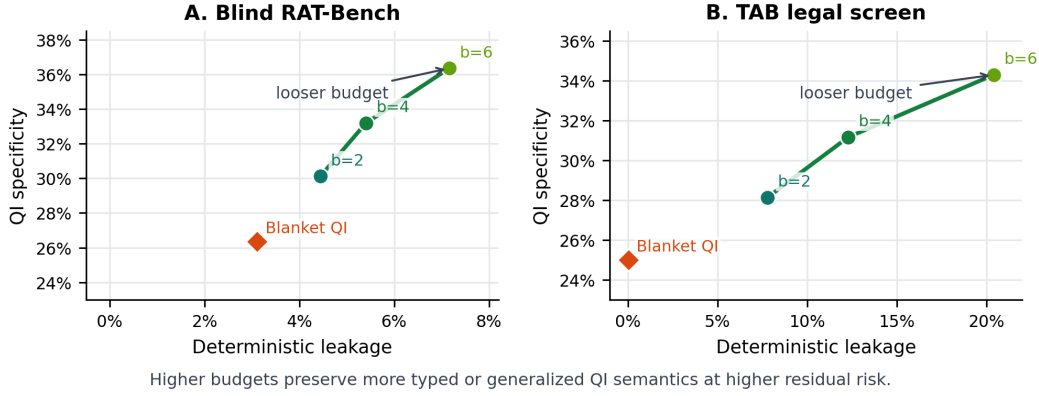


Figure 4: Budget frontier and domain transfer. Deterministic budget sweeps on blind RAT-Bench and TAB show that looser RB-QIG budgets preserve more annotation-derived QI specificity at higher leakage, while blanket QI redaction stays low-specificity.

Table 3: Representative residual failure categories.

| Failure type    | Example cue                              | Why span removal misses it                                          | Possible fix            |
|-----------------|------------------------------------------|---------------------------------------------------------------------|-------------------------|
| Relation cue    | bereavement family-status phrase         | or sensitive status is inferred from a relation, not named directly | relation-aware rewrite  |
| Gender cue      | pronouns or gendered family roles        | sex can be inferred after direct identifiers are removed            | neutralization pass     |
| Cultural cue    | heritage, tradition, or language mention | race or ethnicity may be inferred from ordinary context             | coarser cultural phrase |
| Institution cue | school, employer, or service branch      | education or employment is exposed by a named institution           | institution ladder      |
| Placeholder cue | typed demographic placeholder in context | the placeholder itself can reveal the attribute class               | re-neutral placeholders |

## 6 Limitations and Ethics

The strongest utility result is synthetic and controlled; public utility preservation remains mixed. The blind extractor and deterministic backstop improve coverage, yet blind QI extraction and task-aware rewriting remain open problems. LLM attackers and utility judges are model-dependent and may under- or over-estimate real adversaries or human utility needs. All experiments use synthetic or public benchmark data. We do not attempt to identify real people or infer private attributes about individuals.

## 7 Conclusion

Direct PII removal is an incomplete transformation for LLM-ready sensitive text. RB-QIG reframes de-identification as a risk-budgeted generalization problem: identify quasi-identifiers, score their privacy and utility roles, and choose the least destructive replacements that meet a risk budget. The current evidence supports a clear message. Direct redaction and naive LLM sanitization leave substantial residual inference risk; explicit QI transformation sharply reduces that risk; controlled utility can be preserved better than blanket redaction; and blind, task-aware utility preservation remains the main open problem. Responsible

data enablement should therefore evaluate what capable models can still infer, not only what explicit PII spans were removed.

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